Physics Olympiad L2-26

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Problem 1: **Dielectric Ball**

In this question we study a uniform dielectric ball of radius R and relative permittivity ε in two settings:

Setting 1: The dielectric ball is placed in vacuum in a uniform external electric field of strength $\vec{E}_0$.

- Find the electric field strength $\vec{E}$ inside the ball and the polarization $\vec{P}$ of the dielectric;

A ball that is polarized in this way may be represented as a result of a small rigid shift $d \ll R$ of all positive charges of the dielectric relative to all the negative ones. It can be assumed that there are n atoms per unit volume of the dielectric, each with atomic number Z.

- Justify why the electric field outside of the ball behaves like a dipole at the centre of the ball;
- Find an expression for the total electric potential outside of the ball;
- Find an expression for the total electric field outside of the ball;
- Find an expression for the small shift d;

Setting 2: Three such dielectric balls are fixed on the vertices of an equilateral triangle of side $a \gg R$. The configuration is then placed in a uniform external electric field of strength $\vec{E}_0$ so that the field is perpendicular to one of the sides of the triangle. Assume that $\vec{E}_0$ acts in the plane of the triangle and that enough time has passed for the system to reach equilibrium.

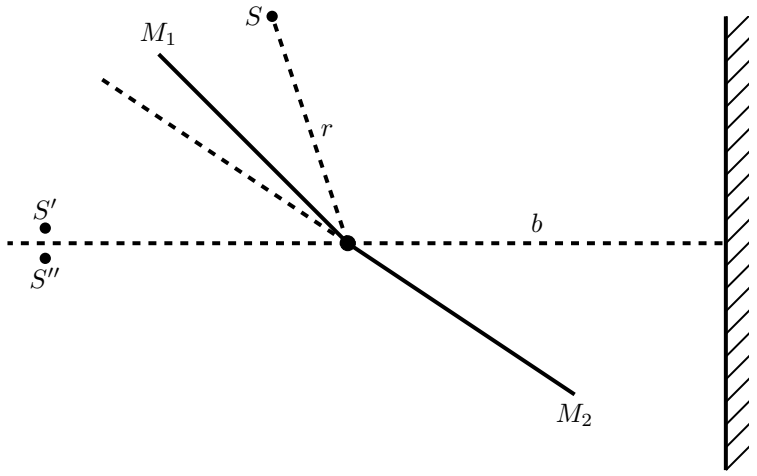
- Taking the potential energy to be zero when the three balls are infinitely separated from each other, find the potential energy of this configuration;

Note: You should not use the dipole trick for the first part of "Setting 1".

Problem 2: **Fresnel Mirrors**

The below figure illustrates the interference experiment with Fresnel mirrors M_1 and M_2. The angle between the mirrors is $\varphi = 12'$, the distances from their intersection line to the narrow slit S and the screen are equal to $r = 10\,\text{cm}$ and $b = 130\,\text{cm}$ respectively. The wavelength of light is $\lambda = 0{,}55\,\mu\text{m}$. Find:

- The width of a fringe on the screen and the number of possible maxima;
- The shift of the interference pattern on the screen when the slit is displaced by $\delta l = 1\,\text{mm}$ along the arc of radius r with centre at the point of intersection of the mirrors;
- At what maximum width δ_{max} of the slit the interference fringes on the screen are still observed sufficiently sharp.



Problem 3: Polarization

What kind of polarization has a plane electromagnetic wave if the projections of the vector $\vec{E}$ on the x and y axes are perpendicular to the propagation direction and are defined as:

- $E_x = E\cos(\omega t - kz)$ and $E_y = E\cos(\omega t - kz)$
- $E_x = E\cos(\omega t - kz)$ and $E_y = E\cos(\omega t - kz + \pi/4)$
- $E_x = E\cos(\omega t - kz)$ and $E_y = E\cos(\omega t - kz + \pi/2)$
- $E_x = E\cos(\omega t - kz)$ and $E_y = E\cos(\omega t - kz + 3\pi/4)$
- $E_x = E\cos(\omega t - kz)$ and $E_y = E\cos(\omega t - kz + \pi)$

Problem 4: Efficiency

Using the Clausius inequality, show that all cycles having the same maximum temperature T_{max} and minimum temperature T_{min} are less efficient compared to the Carnot cycle with the same T_{max} and T_{min}.

Problem 5: The Reduced Compton Wavelength and Compton Scattering

- **The Scale and The Reduced Compton Wavelength**
  In this problem you could assume that the effective mass of a photon is given by the following relation:

$$\boxed{\hbar\omega = m_{eff}c^2}$$

  → Compose an expression for a quantity whose dimension is length, using the velocity of light c, mass of a particle m, and Planck's constant $\hbar$. This expression is the reduced Compton wavelength $\bar{\lambda}$;

  → Evaluate $\bar{\lambda}_e$ for an electron and find the frequency ν of a photon of the same reduced wavelength $\bar{\lambda}_e$;

  → Estimate the spatial scale Δx to which an electron must be confined for its kinetic energy to become comparable to its rest energy. What qualitatively new process becomes possible below this size?

  → In the **Bohr model of hydrogen** the radius a_0 of the ground-state orbit is called the **Bohr radius**. The ratio of the electron's reduced Compton wavelength $\bar{\lambda}_e$ to the Bohr radius defines the **fine-structure constant** α:

$$\boxed{\alpha := \frac{\bar{\lambda}_e}{a_0}}$$

  Derive an expression for α in terms of e, ε_0, $\hbar$ and c, and evaluate its numerical value;

- **Absorption of a Photon**

  → Prove that a free electron cannot absorb a photon completely;

  → A hydrogen atom, initially at rest in its ground state, absorbs a photon and goes to the excited state $n = 2$. Find the exact energy E_γ of the photon and the kinetic energy E_k of the recoiling atom. Hence, evaluate the ratio η which shows by how much E_γ exceeds ΔE:

$$\eta = \frac{|E_\gamma - \Delta E|}{\Delta E}$$

  where ΔE is the excitation energy of the atom in its rest frame. Any approximations you make should be thoroughly reasoned for and explicitly shown to be appropriate;

- **Compton Scattering**

  → Derive the Compton shift $\Delta\lambda$ for scattering on a free electron at rest;

  → Express the energy E'_γ of the scattered photon in terms of the incident photon energy E_γ and the scattering angle φ. Hence, determine the kinetic energy E_k of the recoiling electron;

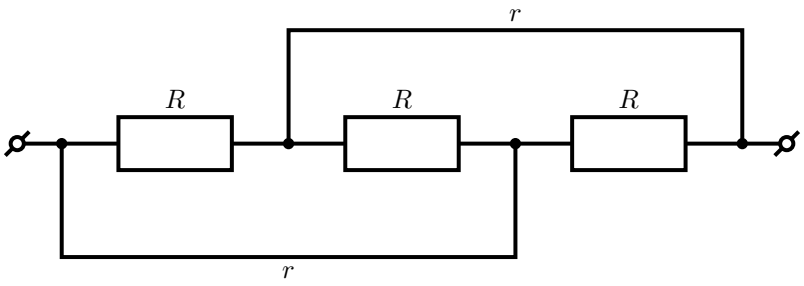
  → Determine the maximum possible wavelength shift and the corresponding scattering angle. Express the result in terms of the reduced Compton wavelength $\bar{\lambda}_e$ and evaluate it numerically;

  → Determine the maximum kinetic energy of the recoiling electron. Examine the limiting cases $E_\gamma \ll m_e c^2$ and $E_\gamma \gg m_e c^2$ and explain their physical significance;

Problem 6: **Resistor Circuit**

A DC source of electromotive force ε and internal resistance R_0 is connected to three identical resistors, each of resistance R, as shown in the figure. The two long connecting branches drawn above and below the three resistors each have resistance r, while all other connecting wires have negligible resistance.

- Obtain an expression for the total thermal power P dissipated in the three resistors;
- Determine the value of R in terms of R_0 for which P is maximized when $r = R_0$ and find P_{max};



Appendix

Useful Mathematical Formulae and Physical Constants

Important: Infinite series that are marked with $(*)$ are only valid under the condition that $|x| < 1$.

Taylor Expansions

$$\sin(x) = x - \frac{x^3}{6} + \frac{x^5}{120} + \mathcal{O}(x^7)$$

$$\cos(x) = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \frac{x^6}{720} + \mathcal{O}(x^8)$$

$$\mathrm{tg}(x) = x + \frac{x^3}{3} + \frac{2x^5}{15} + \mathcal{O}(x^7)$$

$$\arcsin(x) = x + \frac{x^3}{6} + \frac{3x^5}{40} + \mathcal{O}(x^7)$$

$$\mathrm{arctg}(x) = x - \frac{x^3}{3} + \frac{x^5}{5} + \mathcal{O}(x^7)$$

$$\exp(x) = 1 + x + \frac{x^2}{2} + \mathcal{O}(x^3)$$

$$\ln(1+x) = x - \frac{1}{2}x^2 + \mathcal{O}(x^3) \qquad (*)$$

$$(1-x)^{-1} = 1 + x + x^2 + \mathcal{O}(x^3) \qquad (*)$$

$$(1+x)^{-1} = 1 - x + x^2 + \mathcal{O}(x^3) \qquad (*)$$

$$(1+x)^{+\frac{1}{2}} = 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \mathcal{O}(x^3) \qquad (*)$$

$$(1+x)^{-\frac{1}{2}} = 1 - \frac{1}{2}x + \frac{3}{8}x^2 + \mathcal{O}(x^3) \qquad (*)$$

$$(1+x)^{+\frac{3}{2}} = 1 + \frac{3}{2}x + \frac{3}{8}x^2 + \mathcal{O}(x^3) \qquad (*)$$

$$(1+x)^{-\frac{3}{2}} = 1 - \frac{3}{2}x + \frac{15}{8}x^2 + \mathcal{O}(x^3) \qquad (*)$$

Inverse Hyperbolic Functions

$$\mathrm{arsinh}(x) = \ln(x + \sqrt{x^2+1})$$

$$\mathrm{arcosh}(x) = \ln(x + \sqrt{x^2-1})$$

$$\mathrm{artgh}(x) = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right)$$

$$\mathrm{arctgh}(x) = \frac{1}{2}\ln\left(\frac{x+1}{x-1}\right)$$

Indefinite Integrals

$$\int (1-x^2)^{-\frac{3}{2}}\,dx = \frac{x}{\sqrt{1-x^2}} + C$$

$$\int (1+x^2)^{-\frac{3}{2}}\,dx = \frac{x}{\sqrt{1+x^2}} + C$$

$$\int (1-x^2)^{-\frac{1}{2}}\,dx = \arcsin(x) + C$$

$$\int (1+x^2)^{-\frac{1}{2}}\,dx = \mathrm{arsinh}(x) + C$$

$$\int (1-x^2)^{+\frac{1}{2}}\,dx = \frac{\arcsin(x) + x\sqrt{1-x^2}}{2} + C$$

$$\int (1+x^2)^{+\frac{1}{2}}\,dx = \frac{\mathrm{arsinh}(x) + x\sqrt{1+x^2}}{2} + C$$

Useful Physical Constants

Constant	Symbol	Value	Measuring Units
Light speed in vacuum	c	$3{,}000 \times 10^{8}$	m/s
Rest mass of electron	m_e	$9{,}109 \times 10^{-31}$	kg
Rest mass of proton	m_p	$1{,}673 \times 10^{-27}$	kg
Rest mass of neutron	m_n	$1{,}675 \times 10^{-27}$	kg
Rest mass of muon	m_μ	$1{,}884 \times 10^{-28}$	kg
Rest mass of tau-lepton	m_τ	$3{,}168 \times 10^{-27}$	kg
Elementary charge	e	$1{,}602 \times 10^{-19}$	C
Planck's constant	h	$6{,}626 \times 10^{-34}$	Js
Boltzmann's constant	k	$1{,}381 \times 10^{-23}$	J/K
Gravitational constant	γ	$6{,}673 \times 10^{-11}$	$\mathrm{Nm^2/kg^2}$
Dielectric permittivity of vacuum	ε_0	$8{,}854 \times 10^{-12}$	F/m
Avogadro's number	N_A	$6{,}022 \times 10^{23}$	$\mathrm{mol^{-1}}$
Rydberg's constant	R_∞	$1{,}097 \times 10^{7}$	$\mathrm{m^{-1}}$
Magnetic permeability of vacuum	μ_0	$4\pi \times 10^{-7}$	H/m
Universal gas constant	R	8,314	$\mathrm{J/(mol \cdot K)}$